Real-Time Retail Feedback Intelligence

— A GenAI-powered Customer Review Automation Solution —

ChicStyle — a growing women's fashion retail platform

- ChicStyle sees massive spikes in customer activity during festive and holiday sales, leading to a surge in incoming reviews.
- Reviews arrive continuously and vary widely—from positive feedback to urgent issues like fit problems, delivery delays, defects, and sizing concerns.
- Delays in responding during these high-pressure periods can frustrate customers, damage trust, and negatively impact both immediate sales and long-term loyalty.
- To maintain customer satisfaction and protect brand reputation, ChicStyle needs a system that can process large volumes of reviews quickly, accurately, and with business context.

Problem Statement & Objectives

Statement:

- Retailers need an intelligent feedback system that can process high volumes of real-time reviews, detect sentiment, identify relevant products or services, and flag urgent issues.
- Traditional NLP models struggle due to **rule-based approaches and limited training data**, making them less effective for complex feedback.
- In contrast, generative AI understands context, distinguishes multiple opinions, and detects nuanced emotions, enabling more accurate insights and faster responses.

Objectives:

- Analyze and categorize feedback with real-time sentiment detection
- Identify the product/service referenced in each review
- Summarize insights by category and urgency

Impact: Enables faster issue resolution, improves product quality, and converts unstructured feedback into real-time business insights

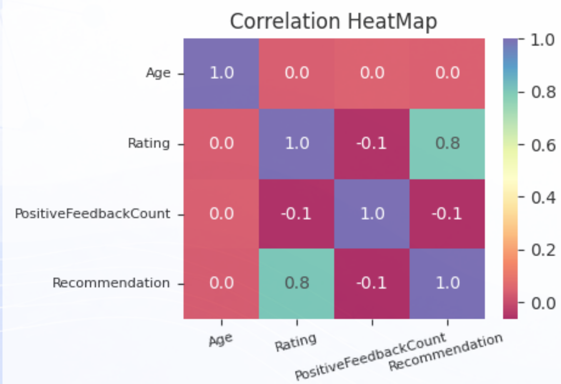
Data Overview

Key dataset fields

- Title: The title of the review.
- Review.Text: The main body of the customer's review text.
- Rating: The product score given by the customer, from 1 (Worst) to 5 (Best).
- Recommended: A binary variable indicating if the customer recommends the product (1 for recommended, 0 for not recommended).
- Positive Feedback Count: The number of other customers who found the review helpful (Positive Integer).

Interesting statistics

Correlation Heatmap



Frequent words(WordCloud)

EDA Findings & Insights

- **Customer Ratings:** Highly skewed toward positive, with **75%+ rated 5**, but lower ratings carry more informational value.
- **Key Drivers: Fit/size and appearance (color/style)** are the main factors influencing satisfaction—especially for Dresses and Tops.
- **Category Performance:**
  - Strong: Intimate, Bottoms, Jackets (functionality-driven)
  - Weak: Dresses, Tops, Trend (appearance-driven expectations)
- Customer Behavior Insight: Reviews with low ratings and no recommendations are perceived as more helpful, indicating higher decision impact.
- **Modeling Implication:** Presence of **mixed sentiment within single reviews** highlights the need for advanced AI (e.g., generative models) to capture nuanced feedback accurately.

Data Pre-Processing

Review Text is the key input for the LLM

- Ignore missing reviews (low %; no response expected) ✅ *(used in this project)*
- Alternative: use default input (“No comment”) via prompt logic

LLM advantages:

- Handles misspellings & incomplete sentences
- Minimal preprocessing needed

Data handling:

- Missing categorical fields (Division, Department, ClassName) → excluded
- No preprocessing needed for Rating, Recommendation, PositiveFeedbackCount (no missing values)

Prompting Techniques

- Zero-Shot prompt
  - Define model role and required outputs
  - Enforce strict JSON format
  - Specify allowed labels
  - No examples provided

- Few-Shot prompt
  - Add 3–5 examples before the task
  - Improves consistency for tricky cases: Fit vs Quality, mixed sentiment, primary vs secondary issues
  - Includes field-specific rules

- Chain-of-Thought prompt
  - Uses structured internal reasoning
  - Does not output reasoning steps
  - Outputs final JSON only
  - Improves quality through step-by-step decision logic

Each prompt level builds on the previous one.

Retrieval-Augmented Generation (RAG)

- Why RAG
  - LLMs rely only on prompt input → can miss business context.
  - Improves consistency across similar customer issues and feedbacks.
  - Reduces hallucination and unsupported responses.
  - Keeping the instructions clear and structured.
  - Aligns outputs with real-world patterns and expectations.
- How RAG Improves Prompting Performance
  - Retrieves top-k similar past cases before generation.
  - Injects grounded context into the prompt.
  - Guides the model toward more accurate classification and responses.
  - Enhances CoT reasoning with real examples.
- Result
  - Higher accuracy
  - Better consistency
  - More reliable, production-ready outputs

Note: *In the proof-concept experiment, we use manually refined predictions (via ChatGPT) as ground truth for 20 reviews. The LLM judge compares outputs to these references based semantically. Although a small and curated dataset, it demonstrates clear improvements across metrics and overall performance.*

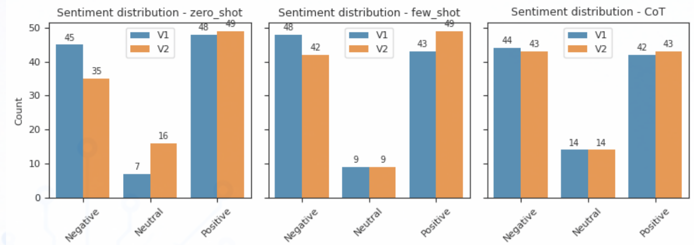
Model Evaluation – Zero-Shot/Few-Shot/CoT/RAG+COT

Great Learning
POWER AHEAD

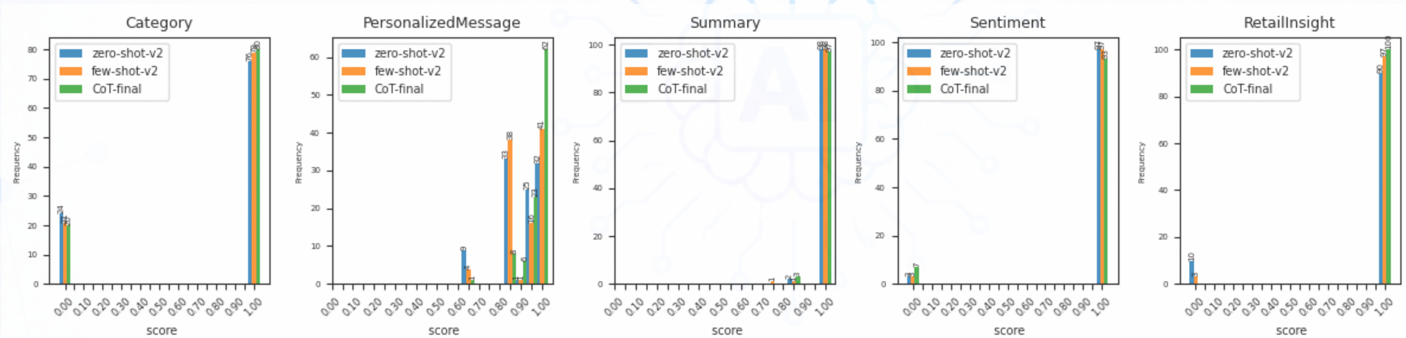
- **Prompt performance average scores**

Prompt	*V1*	*V2*	*Final*
RAG+CoT	x	x	0.960
CoT	0.784	0.932	0.939
Few-Shot	0.780	0.929	x
Zero-Shot	0.772	0.906	x

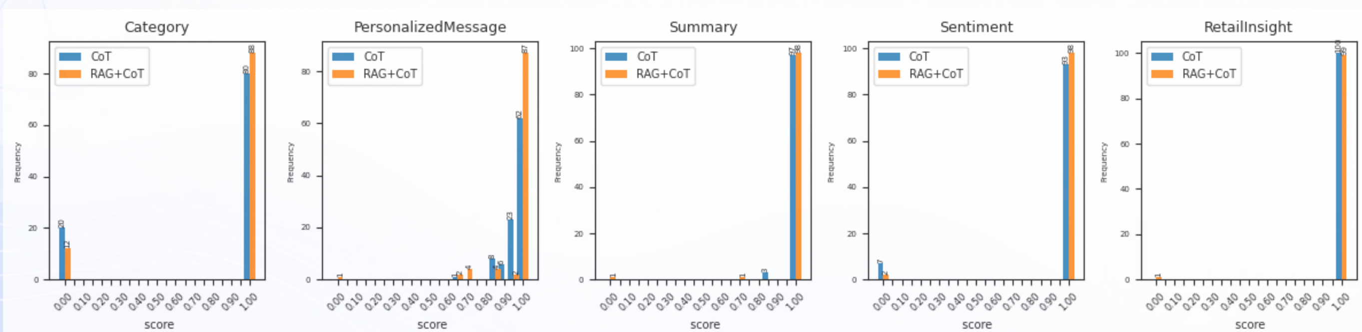
- **Sentiment Prediction distribution**



- **LLM-judge score distributions by output field (Zero-Shot/Few-Fhot/CoT)**



- **LLM-judge score distributions by output field (CoT/RAG+CoT)**



Findings & Insights – Zero-Shot/Few-Shot/CoT/RAG+COT

Findings

- V2 > V1 across all techniques (better structured rules)
- Few-shot >> Zero-shot; CoT > Few-shot
- Scores (Category, Summary, Retail Insight) concentrated at high end
- CoT shows stable sentiment (less prompt drift)
- RAG + CoT achieves highest performance

Insights

- Best overall: **RAG + CoT**, followed by CoT
- CoT improves reasoning (infer → compare → decide → explain), reducing errors
- RAG adds grounding → less hallucination, more consistency
- RAG + CoT = most reliable and consistent results

CoT Review Intelligence Engine

- Improves accuracy for mixed sentiment, fit, quality, delivery, urgency, and responses

RAG-Grounded Response Layer

- Uses past cases + product data for grounded, consistent responses
- Reduces hallucination; improves traceability

Agentic Framework + Harness

- Orchestrated LLM agents with automated evaluation for quality and continuous improvement

Real-Time Dashboard

- Urgent issues, top problems, category insights
- Response status & sentiment trends

Human-in-the-Loop

- Escalates high-urgency reviews to support teams

Implementation Recommendations

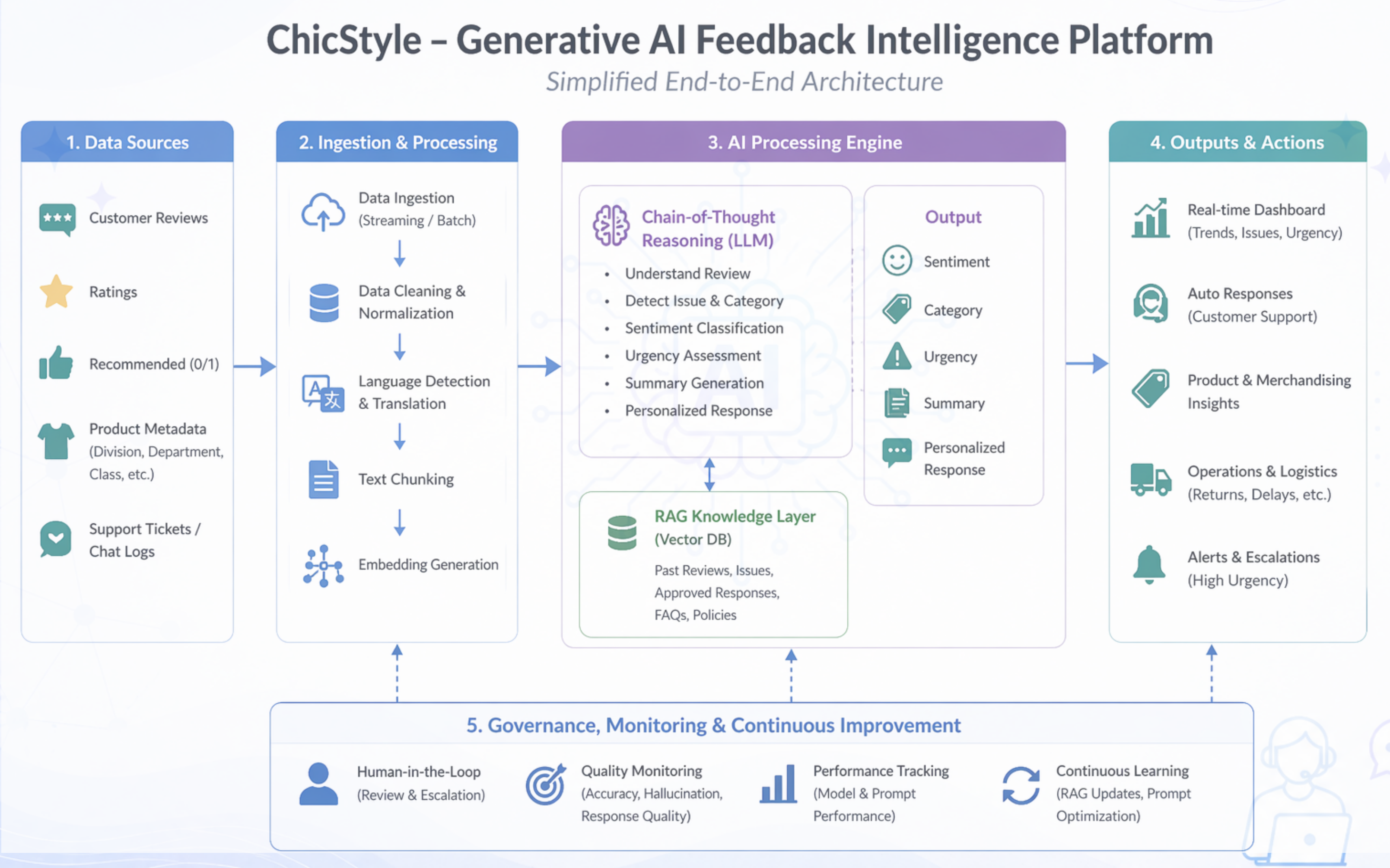
Short-Term (3–6 Months)

- Deploy CoT-based classification (sentiment, category, urgency, summary) → faster, more accurate triage
- Add RAG-grounded responses (historical + product data) → consistent, brand-safe communication
- Launch real-time dashboard (sentiment, urgency, issues) → quicker decisions

Long-Term (6–12 Months)

- Integrate with CRM & auto-route high-urgency issues → faster resolution
- Build product trend detection (fit, fabric, sizing, delivery) → reduce returns
- Establish continuous learning & governance → ongoing improvement & risk reduction

Solution Architecture



Estimated Business Benefits & Costs

Estimated Benefits (Pilot & Early Scale)

- 30–50% less manual review effort
- 2–3× faster detection of negative/urgent issues
- More consistent categorization & summaries
- Earlier detection of product issues (fit, quality, delivery)
- Better customer experience with faster responses

Basis of Estimates

- 5–10 min manual review → seconds with CoT
- 10K–100K reviews/month
- Based on typical support workflows (excludes future RAG/CRM gains)

Structure (Pilot Phase)

Development Costs (One-time): ~$6K–14K

Monthly Ops: ~$400–1,000

- LLM API: $200–500
- Cloud: $100–300
- Maintenance: $100–200

Cost per Review: ~$0.002–0.01 → highly scalable

Risks & Mitigation

- Model Misclassification and Hallucination Risk
  - Incorrect sentiment and urgency → wrong action
  - Hallucinated summary and personalized message.
  - Mitigation:
    - ✓ Model temperature control.
    - ✓ RAG implementation to ground answers based on historical curated responses.
    - ✓ human-in-the-loop for high urgency cases
- Cost Scalability
  - High API usage during peak seasons
  - Mitigation: hybrid prompt strategy
- Integration Complexity
  - Connecting with CRM, support, product systems
  - Mitigation: phased rollout

Conclusion

- When fixed rules and limited data fall short, GenAI handles complexity and ambiguity better than traditional ML

- **Trade-off:** Higher cost due to LLM usage (e.g., API quotas and service fees)

THANK YOU

Any Questions?